REVIEW - EXAM 3

BLAKE FARMAN

Lafayette College

Name: Solutions

1. Problems

For each of the following problems, decide which of the tests listed above is appropriate and use it to show that the given series converges or diverges. **Clearly indicate the test you used. If the test has hypotheses, clearly indicate why the series fits the hypotheses of the test.**

1 (5 Points)**.** Determine whether the series $\sum_{n=0}^{\infty} \frac{5(3^n) + 2^{n+1}}{6^n}$ converges or diverges. If the series converges, find the sum of the series.

$$\sum_{n=0}^{\infty} \frac{5(3^n)+2^{n+1}}{6^n} = \sum_{n=0}^{\infty} 5\left(\frac{3}{6}\right)^n + \sum_{n=0}^{\infty} 2\left(\frac{2}{6}\right)^n = \sum_{n=0}^{\infty} 5\left(\frac{1}{2}\right)^n + \sum_{n=0}^{\infty} 2\left(\frac{1}{3}\right)^n$$

$$= \frac{5}{1-\frac{1}{2}} + \frac{2}{1-\frac{1}{3}} = \frac{5}{1/2} + \frac{2}{2/3} = 10+3 = \boxed{13}$$

2 (5 Points)**.** Determine whether the series $\sum_{n=0}^{\infty} \frac{e^n}{e^n + n}$ converges or diverges.

Diverges by the nth Term Test for Divergence:

$$\lim_{n\to\infty} \frac{e^n}{e^n+n} \overset{L'H}{=} \lim_{n\to\infty} \frac{e^n}{e^n+1} \overset{L'H}{=} \lim_{n\to\infty} \frac{e^n}{e^n} = 1 \neq 0.$$

3 (5 Points)**.** Determine whether the series $\sum_{n=1}^{\infty} \frac{\ln(n)}{n}$ converges or diverges.

Diverges by comparison with the Harmonic Series:

$$e < n \Rightarrow \ln(e) = 1 < \ln(n)$$

$$\Rightarrow \frac{1}{n} < \frac{\ln(n)}{n}$$

4 (10 Points)**.** Determine whether the series $\sum_{n=1}^{\infty} \frac{n+1}{n^2\sqrt{n}}$ converges or diverges.

Convergent by L.C.T. with $\sum_{n=1}^{\infty} \frac{1}{n^{3/2}}$

$$\lim_{n\to\infty} \frac{n+1}{n^{5/2}} \frac{n^{3/2}}{1} = \lim_{n\to\infty} \frac{n+1}{n} = \lim_{n\to\infty} 1+\frac{1}{n} = 1+0 = 1$$

5 (10 Points)**.** Determine whether the series $\sum_{n=1}^{\infty} (-1)^n \frac{n^2(n+2)!}{n!3^{2n}}$ converges conditionally, converges absolutely, or diverges. Converges absolutely by Ratio Test:

$$\lim_{n\to\infty} \left| \frac{(n+1)^2(n+3)!}{(n+1)!\,3^{2n+2}} \frac{n!\,3^{2n}}{n^2(n+2)!} \right| = \lim_{n\to\infty} \frac{(n+1)^2}{n^2} \frac{(n+3)!}{(n+2)!} \frac{n!}{(n+1)!} \frac{3^{2n}}{3^{2n+2}}$$

$$= \lim_{n\to\infty} \frac{(n+1)^2}{n^2} \frac{(n+3)}{n+1} \frac{1}{9} = \lim_{n\to\infty} \frac{(n+1)(n+3)}{9n^2} = \lim_{n\to\infty} \frac{n^2+4n+3}{9n^2} = \frac{1}{9} < 1$$

6 (15 Points)**.** Determine whether the series $\sum_{n=1}^{\infty} \left(\frac{4n+3}{3n-5}\right)^n$ converges or diverges.

Diverges by the Root Test:

$$\lim_{n\to\infty} \sqrt[n]{\left(\frac{4n+3}{3n-5}\right)^n} = \lim_{n\to\infty} \frac{4n+3}{3n-5} = \frac{4}{3} > 1.$$

7 (15 Points)**.** Determine whether the series $\sum_{n=1}^{\infty}(-1)^n \frac{2n}{4n^2+1}$ converges conditionally, converges absolutely, or diverges.

Converges conditionally by A.S.T

$f(x) = \frac{2x}{4x^2+1} \Rightarrow f'(x) = \frac{2(4x^2+1) - 2x(8x)}{(4x^2+1)^2} = \frac{-8x^2+2}{(4x^2+1)^2} < 0$ for $x \geq 1$.

$\Rightarrow \frac{2(n+1)}{4(n+1)^2+1} = f(n+1) \leq f(n) = \frac{2n}{4n^2+1}$ but $\sum_{n=1}^{\infty} \frac{2n}{4n^2+1}$ diverges by L.C.T.

with the Harmonic Series:

$$\lim_{n\to\infty} \frac{2n}{4n^2+1}\left(\frac{n}{1}\right) = \lim_{n\to\infty} \frac{2n^2}{4n^2+1} = \frac{2}{4} = \frac{1}{2}.$$

8. Determine whether the series $\sum_{n=1}^{\infty} \sin\left(\frac{1}{n}\right)$ converges or diverges.

Diverges by L.C.T. with Harmonic Series

$$\lim_{n\to\infty} \frac{\sin(\frac{1}{n})}{\frac{1}{n}} \overset{L'H}{=} \lim_{n\to\infty} \frac{\cos(\frac{1}{n})\frac{d}{dn}(\frac{1}{n})}{\frac{d}{dn}(\frac{1}{n})}$$

$$= \lim_{n\to\infty} \cos(\tfrac{1}{n})$$

$$= \cos(0) = 1 > 0.$$

9. Find the radius and interval of convergence for the power series

$$\sum_{n=0}^{\infty} \frac{3^n x^n}{n!}.$$

$$\lim_{n\to\infty} \left| \frac{3^{n+1} x^{n+1}}{(n+1)!} \frac{n!}{3^n x^n} \right| = \lim_{n\to\infty} \frac{3|x|}{n+1} = 0 < 1$$

for all x. Therefore

$$\sum_{n=0}^{\infty} \frac{3^n x^n}{n!} \text{ converges } (-\infty, \infty).$$

10 (25 Points)**.** Find the Maclaurin series for the function

$$f(x) = \frac{1}{(1-x)^3}.$$

$$g(x) = (1-x)^{-1} = \sum_{n=0}^{\infty} x^n \text{ on } (-1,1)$$

$$g'(x) = -(1-x)^{-2}(-1) = (1-x)^{-2} = \sum_{n=0}^{\infty} n x^{n-1} \text{ on } (-1,1)$$

$$g''(x) = -2(1-x)^{-3}(-1) = 2(1-x)^{-3} = \sum_{n=0}^{\infty} n(n-1)x^{n-2} \text{ on } (-1,1).$$

$$\Rightarrow \frac{1}{(1-x)^3} = \frac{1}{2} \sum_{n=0}^{\infty} n(n-1)x^{n-2} = \sum_{n=2}^{\infty} \frac{n(n-1)}{2} x^{n-2} \text{ on } (-1,1)$$

11 (25 Points)**.** Find the Maclaurin series for $x\ln(1+2x)$.

$$f(x) = \ln(1+x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{n+1}}{n+1} \quad \text{on } (-1,1)$$

$$\Rightarrow \ln(1+2x) = f(2x) = \sum_{n=0}^{\infty} \frac{(-1)^n (2x)^{n+1}}{n+1}$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n 2^{n+1} x^{n+1}}{n+1} \quad \text{for } -1 < 2x < 1$$

$$\Leftrightarrow -\tfrac{1}{2} < x < \tfrac{1}{2}$$

$$\Rightarrow x\ln(1+x) = x \sum_{n=0}^{\infty} \frac{(-1)^n 2^{n+1} x^{n+1}}{n+1}$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n 2^{n+1} x^{n+2}}{n+1} \quad \text{on } \left(-\tfrac{1}{2}, \tfrac{1}{2}\right).$$

12 (25 Points)**.** Find the Maclaurin series for

$$\ln\left(\frac{1+x}{1-x}\right)$$

$$\ln\left(\frac{1+x}{1-x}\right) = \ln(1+x) - \ln(1-x)$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n x^{n+1}}{n+1} - \sum_{n=0}^{\infty} \frac{(-1)^n (-x)^{n+1}}{n+1}$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n x^{n+1}}{n+1} + \sum_{n=0}^{\infty} \frac{(-1)^{n+1}(-1)^{n+1} x^{n+1}}{n+1}$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n x^{n+1}}{n+1} + \sum_{n=0}^{\infty} \frac{x^{n+1}}{n+1}$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n x^{n+1} + x^{n+1}}{n+1}$$

$$= \sum_{n=0}^{\infty} \frac{((-1)^n + 1) x^{n+1}}{n+1}$$

$$= \sum_{n=0}^{\infty} \frac{2x^{2n+1}}{2n+1} \quad \text{on } (-1, 1)$$

13. Determine whether the series

$$\sum_{n=1}^{\infty}(-1)^n \frac{2n}{4n^2+1}$$

converges conditionally, converges absolutely, or diverges.